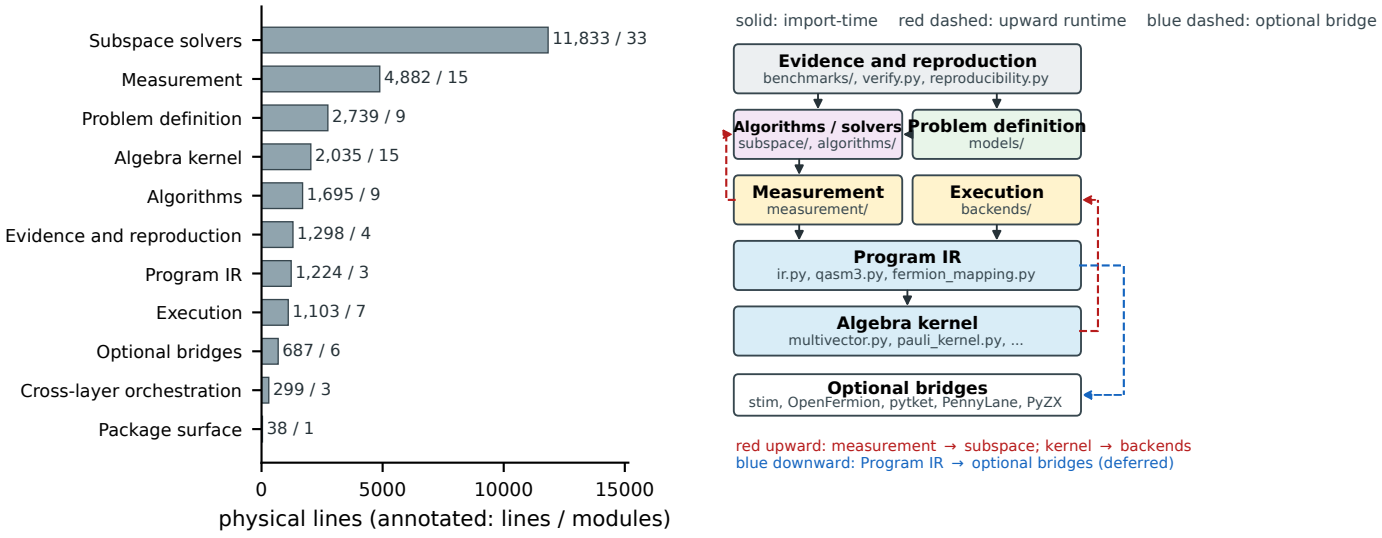
(a) Declared-layer size
Subspace solvers
Measurement
Problem definition
Algebra kernel
Algorithms
Evidence and reproduction
Program IR
Execution
Optional bridges
Cross-layer orchestration
Package surface
11,833 / 33
4,882 / 15
2,739 / 9
2,035 / 15
1,695 / 9
1,298 / 4
1,224 / 3
1,103 / 7
687 / 6
299 / 3
38 / 1
0
5000
10000
15000
physical lines (annotated: lines / modules)
(b) Selected dependency edges (partial)
solid: import-time
red dashed: upward runtime
blue dashed: optional bridge
Evidence and reproduction
benchmarks/, verify.py, reproducibility.py
Algorithms / solvers
subspace/, algorithms/
Problem definition
models/
Measurement
measurement/
Execution
backends/
Program IR
ir.py, qasm3.py, fermion_mapping.py
Algebra kernel
multivector.py, pauli_kernel.py, ...
Optional bridges
stim, OpenFermion, pytket, PennyLane, PyZX
red upward: measurement → subspace; kernel → backends
blue downward: Program IR → optional bridges (deferred)